

(P108) Local Control and Toxicity of Reirradiation With Pulsed Low-Dose-Rate Radiation



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Purpose: To evaluate the efficacy and toxicity of reirradiation using pulsed low-dose-rate (PLDR) technique.

Methods/Materials: We evaluated patients treated with reirradiation using PLDR between 2009-2016 at a single institution. Toxicity was graded based on CTCAE4.0 and local control was assessed using RECIST1.1. Kaplan-Meier estimation and Cox proportional hazard model were used for univariate analysis (UVA). Multivariate analysis was not performed due to the small number of events.

Results: Thirty-nine patients were treated to 41 sites with PLDR reirradiation, with a median follow-up of 8.8 months (range 0.5-64.7 months). Targets included 25 thoracic, 3 abdominal, and 13 pelvic. Thirty-six were symptomatic, including 20 with pain. The intent was palliative in 87%. The median time between the first course of radiation and reirradiation was 26.2 months. The median doses of the first and second courses of radiation were 50.4 Gy and 50 Gy. Five patients (13%) received concurrent systemic therapy.

Of the 39 patients, 9(23%) developed G2+ acute/subacute toxicity, most commonly skin (5/9). None developed G4+ acute/subacute toxicity. The only late G2+ toxicity was 1 patient with G2 skin toxicity. On UVA, toxicity was not significantly associated with dose of the first course of radiation or reirradiation, time interval between the two courses of radiation, or reirradiation site.

Of the 41 sites treated with PLDR, 32 were assessed for local control. Local control rates were 85% at 6 months and 76% at 12 months. Local control was not significantly associated with the dose of reirradiation, reirradiation site, or concurrent systemic therapy on UVA. Of the 36 symptomatic sites, 6(15%) achieved complete symptomatic relief and 19(46%) achieved partial relief. Pain improved in 14/20 (70%) patients.

Conclusion: Based on this retrospective study, reirradiation with PLDR is effective and well-tolerated. Assessment of late toxicity was limited by the relatively short follow up in this cohort.

(P109) Factors Associated With Increase in Fatigue in Prostate Cancer (PC) Patients Undergoing External Beam Radiation Therapy (EBRT)



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Purpose/Objectives: Fatigue is a commonly-reported symptom among cancer patients undergoing EBRT. Despite high penetrance of fatigue and its adverse impact on quality-of-life in this population, the biology of radiation-related fatigue is unclear. The underlying disease- and treatment-related factors influencing fatigue are also poorly understood. This study aims to characterize fatigue development and potential drivers of increased fatigue in PC patients during EBRT.

Materials/Methods: We identified a cohort of 318 patients with non-metastatic PC undergoing a 7-9 week course of EBRT enrolled on our institutional patient registry. Patient fatigue scores (range 0-3, 3 = fatigue unrelieved by rest, limiting self-care) were prospectively recorded at on-treatment visits by providers using NCI-CTCAE v4.0 criteria. For each patient, we recorded clinical and demographic factors including age, race, treatment modality, disease stage, Gleason score, baseline PSA level, daily treatment time, fractions delivered, dose delivered, and concurrent medication use. Non-parametric-means testing was employed to identify significant differences in provider-recorded fatigue levels across the treatment course by classification category.

Results: Consistent with clinical observations, PC patients experienced increasing fatigue levels as the EBRT course progressed, with mean scores

of 0.15 at week 1 (95% CI 0.12-0.18) and 0.59 at week 9 (95% CI 0.54-0.65). No demographic-, disease- or EBRT-related factors correlated with increased fatigue. Fatigue levels were significantly increased in active users of anti-depressants ($p = 0.02$), anti-emetics ($p=0.01$), anti-neoplastics ($p=0.01$), miscellaneous endocrine agents ($p=0.02$), and urinary anti-spasmodics ($p=0.001$) throughout treatment.

Conclusion: The phenomenon of increased fatigue during EBRT was confirmed in this retrospective analysis of a PC patient registry with prospectively-collected fatigue scores. In contrast to patient- and disease-specific factors, the concurrent use of certain medications classes appeared to significantly associate with increased fatigue reported during EBRT. Continued investigation is needed to further elucidate clinical drivers and biological underpinnings of increased fatigue to guide potential interventions.

(P110) A Modified Edmonton Symptom Assessment Scale for Symptom Clusters in Radiation Oncology Patients



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Objective: There is increasing recognition of the importance of assessing patient-reported symptoms as part of clinical care. Recent randomized trials of clinic-based symptom monitoring have found that it is associated with reduced symptom severity, better quality of life, and reduced hospitalization and emergency room visits. Our clinic established use of a modified Edmonton Symptom Assessment Scale (ESAS) in August 2015. All patients presenting for radiation oncology (RO) services complete the form at each clinic visit. As these patients often experience multiple symptoms that affect quality of life and outcomes, recent symptom management literature focuses on identifying symptom clusters.

Methods: Symptom clusters are defined by two or more symptoms that are interrelated and occur simultaneously with a high degree of predictability. An unselected sample of 916 surveys was assessed statistically using principal component analysis (PCA) with varimax rotation on the symptoms to determine independent clustering. Correlative statistics were performed using Pearson's correlation by SAS 9.4; a correlation value of 0.60 was considered significant.

Results: Four clusters were identified: {tired, drowsiness}, {nausea, lack of appetite}, {overall & spiritual well-being}, and {depression, anxiety}. These accounted for 46%, 9.2%, 7.6% and 7% of total variance respectively. Internal reliabilities using Cronbach's alpha coefficient were 0.87, 0.7, 0.82, and 0.87, respectively. The most frequent write-in item was itchiness, present in 24% of the 148 patients responding.

Conclusions: In this first assessment of a large RO population, several symptom clusters present. Fatigue and drowsiness represents a major symptom cluster. Itchiness may be an under-recognized symptom. In addition to being able to use these data for observational analyses, we have been using them to cue clinical inventions that support patients' specific needs. Ultimately we plan to generate a dashboard at the provider level so that patients requiring and receiving special symptom management will be better tracked.

(P111) Long Term Visual Outcomes for Ocular Melanoma Patients Treated With I-125 Brachytherapy Eye Plaques



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Purpose: To report our experience with respect to toxicity with long-term follow-up of ocular melanoma patients treated with COMS design custom I-125 low dose rate brachytherapy eye plaques.

Methods: A retrospective chart review was performed for 127 ocular melanoma patients with follow-up visual acuity data who were treated with